

Harshitha Katta

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Education

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat B.Tech in Computer Science and Engineering (2024 – Present)

- GPA: 7.81/10.0

Experience/Academic Projects

Obstacle Detector for Visually Impaired(Hardware Group Project)

[Youtube Link](#) 

Technologies: Arduino, Ultrasonic Sensor (HC-SR04), Embedded C, AI-based Motion Analysis

- Developing an intelligent obstacle detection system using ultrasonic sensors to assist visually impaired individuals in safe navigation, processing real-time distance data for reliable low-latency obstacle detection.
- Implemented a real-time motion analysis algorithm to classify static and moving obstacles using sequential distance readings and threshold logic, integrating microcontroller-controlled vibration and buzzer modules for immediate tactile and audio feedback.

Full-stack Generative AI web application (Group Project)

Ongoing

- Collaborated with a team Building a full-stack Generative AI web application using Next.js (React), TypeScript, and a Node.js backend integrated with LangChain for LLM-based text generation.
- Implemented modular API routes and background workers for asynchronous AI processing with secure API handling and environment-based configuration.

Zoo Management System (C++ OOP Group Project)

[GitHub Link](#) 

- Designed and developed a Zoo Management System App in C++ by applying Object-Oriented Programming (OOP) principles including encapsulation, inheritance, polymorphism, and abstraction, implementing structured class hierarchies for animals (Animal, Bird, Mammal, Reptile), staff (Staff, Caretaker, Veterinarian), and enclosures to ensure modular and reusable design.
- Built an interactive GUI using Qt with dialog windows and main management screens, organizing the project into well-structured header and source files to enhance maintainability, scalability, and code readability.

Clubs and Activities

Junior Developer at Google Developer Group On Campus - GDGC

- Serving as a Junior Developer at GDGC, participating in coding challenges, technical workshops, build small projects, teamwork skills and knowledge-sharing sessions organized by the club.

Coding Club Member– Nexus(SVNIT)

- Participating enthusiastically in coding challenges, and knowledge-sharing sessions organized by the club.

NSS Volunteer- Cultural committee

- Participated in NSS activities and college cultural events, showcasing teamwork, leadership, and cultural awareness.

Technologies

C, C++, Python, SQL, JavaScript — HTML, CSS, React.js, Next.js — Generative AI, LangChain — OOP, Modular Programming